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, and diarrhea³.

ms after COVID-19 infection may be pment of new health conditions, as an example of the pelvic floor muscles (PFM) dysfunctions. Previous literature suggested that the decrease of muscle strength after COVID-19 infection might affect the lower urinary tract function⁴. One of the most common complains of PFM dysfunctions is the presence of urinary incontinence (UI), described as an involuntary loss of urine⁵. There are previous studies that already hyphothetized that the worsening of urinary symptoms may be related with the chronicity of cough⁶, suggesting that it increases the intra-abdominal pressure and trigger the pelvic floor muscle (PFM) fatigue⁷, resulting in urinary leakage. However, there is no previous evidence that aimed to investigate the relationship between the new-onset of urinary complains and presence of symptoms after COVID-19 infection.

Considering that one of the most common symptoms of post-COVID infection (i.e., cough) is highly correlated with the presence of urinary complains^{4,6,7}, the association between the new-onset of UI should be investigated, as it can adversely influence the quality of life and can reduce women's physical, psychological, and social well-being⁸. Thus, implications about the new-onset of UI might clarify for health professionals which strategies should be taken in order to treat the urinary symptoms, especially because many urogynecological procedures and appointments were canceled during quarantine, as procedures related to severe cases of COVID-19 infection were prioritized⁹.

Therefore, this study aimed to determine if the most common symptoms after COVID-19 infection (i.e., cough, difficulty in breathing, fatigue, muscle pain, loss of movement, tingling, and diarrhea) may explain the presence of new onset of urinary symptoms in female survivors. Furthermore, it analyzed the impact of COVID-19 symptoms on incontinent women's quality of life.

METHODS

Study design

This is a cross-sectional study developed in Brazil and reported according to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines¹⁰. As determined by Resolution 466/2012 of the Brazilian National Health Council, this study was approved by the Institutional Ethics Committee. All participants agreed to participate by reading the online informed consent form and clicking "I agree to participate

if they were Brazilian women aged 18 or older, that were residing in the national territory, who recovered from COVID-19 infection within three months before the data collection and had developed UI during or after the COVID-19 infection, and those who did not have UI before the COVID-19 infection. Participants who did not complete the questionnaire, pregnant women, and women who treated urinary symptoms after COVID-19 infection were excluded.

Procedures

The data collection was carried online by using the Google Forms platform, from January to April 2022. Participants answered a semi-structured and self-administered questionnaire elaborated by the research team to assess the participants' sociodemographic characteristics and information regarding COVID-19 infection. All data collected was self-reported. The questionnaire included questions about personal information (i.e., age, nationality, marital status, and others) and questions about the presence of the most common symptoms after COVID-19 infection (i.e., cough, difficulty in breathing, fatigue, muscle pain, loss of movement, tingling, and diarrhea)¹¹. The presence of cough was classified as acute when it was present for a period of up to three weeks, subacute in cases where it persists between three and eight weeks, and chronic when it last more than eight weeks¹².

To assess the history of UI, the participants answered a question about the onset of the symptom by signaling if UI have started before COVID-19 infection, during the infection or during the recovery of the infection. To evaluate the presence of UI, women answered the Brazilian Portuguese version of the Three Incontinence Questionnaire (3IQ-Br)¹³, a brief and self-administered questionnaire with three questions that distinguishes UI types (stress urinary incontinence, urgency urinary incontinence, and mixed urinary incontinence). This instrument has substantial reliability in the test-retest ($\kappa=0.66$), presented an almost perfect agreement with the King's Health Questionnaire (KHQ) and the Questionnaire for Urinary Incontinence Diagnosis (QUID-Br) to diagnose UI ($\kappa>0.8$) and it has good accuracy in distinguishing women with UI (AUC 0.83, 95% confidence interval [CI] 0.78 to 0.87)¹³.

Participants answered the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF) questionnaire that measures UI impact on women's quality of life¹⁴. It consists of four items and total score ranges from 0-21: the higher the score, worst is the impact of UI severity in quality of life¹⁴. This instrument is translated, adapted, and validated in Brazil. It has a high internal consistency (Cronbach's α 0.88) and moderate to strong test-retest (Kappa index ranged from 0.69 to 0.91, and

& Lemeshow 0.63].

Table 2 shows the odds ratios for SUI, UUI and MUI [(represented by Exp (B))] according to the presence of symptoms after COVID-19 infection. For example, symptoms such as cough, muscle pain, and loss of movement predicted SUI, demonstrating that these symptoms increase the chances of reporting SUI by 1.53, 1.41, and 2.15 times, respectively. Cough and diarrhea increase the chances of women reporting UUI by 0.58 and 1.75 times, respectively. Difficulty breathing, muscle pain, tingling, and diarrhea were predictive of MUI.

Table 3 demonstrated an effect of post-COVID-19 symptoms on the ICIQ-SF score ($F(4;874)=28.24$, $p<0.001$; R^2 adjusted=0.11). It seems that the presence of post-COVID-19 symptoms adversely impacted the quality of life of women with UI. The beta results indicate that muscle pain severely impacted the quality of life of women with UI, explaining 6.3% of the outcome. Other symptoms

Table 1: Sociodemographic characteristics, presence of urinary incontinence, and post-COVID-19 symptoms.

Variables	N	%
Marital status		
Single	188	21.4
Married/stable union	593	67.5
Divorced	76	8.6
Widow	22	2.5
Education		
Primary	105	11.9
High school	415	47.2
College	359	40.8
Post-COVID-19 symptoms		
Coughing	186	21.2
Breathing difficulty	181	20.6
Fatigue	576	65.5
Muscle pain	488	55.5
Loss of movement	37	4.2
Tingling	237	27.0
Diarrhea	139	15.8
Three incontinence questionnaire (3IQ-Br)		
Stress Urinary Incontinence	213	24.2
Urge Urinary Incontinence	135	15.4
Mixed Urinary Incontinence	192	17.4
Onset of UI symptoms		
During COVID-19 infection	116	13.2
Recovery of COVID-19	424	24.0
Continent women	339	38.6

RESULTS

A total of 879 women were included. Participants had a mean age of 40.4 years (SD 9.4), body mass index (BMI) of 30.1 (SD 6.6) kg/m², and a mean ICIQ-SF score of 5.3 points (SD: 5.8), which does not represent a severe impact of UI on quality of life. Table 1 shows the sociodemographic characteristics, presence of symptoms after COVID-19 infection, and UI (according to the 3IQ-Br questionnaire).

The presence of cough, muscle pain and loss of movement had a statistically significant impact on SUI, predicting 75.7% of cases in total [$X^2(3)=17.34$, $p<0.001$; Nagelkerke $R^2=0.029$, Hosmer & Lemeshow 0.85]. Regarding UUI, the presence of cough and diarrhea were statistically significant associated with UUI, adequately predicting 84.6% of cases in total [$X^2(2)=8.32$, $p<0.01$; Nagelkerke

significant impact and therefore were

of cough was not statically significant considering the impact on the quality of life of women with UI, the duration of the cough was investigated to verify whether it interferes with this outcome. A multiple linear regression analysis was performed to investigate whether the duration of cough affects the quality of life of women with UI. The results showed a significant effect on the ICIQ-SF score ($F(1;876)=13.0$, $p=0.001$; R^2 adjusted=0.027). It was possible to observe that the symptom of cough lasting more than eight weeks impacted the quality of life of women with UI, explaining 2.7% of the outcome. The presence of cough lasting three weeks also decreases the quality of life¹⁵. Table 4 shows the predictor coefficients.

were performed. However, due to the slight variation in means between groups with and without UI, it was determined that those results would be presented in table 5. After controlling for confounding factors, a positive and significant correlation between quality of life and all COVID-19 symptoms analyzed was found (Table 6). Additionally, it seems that only cough lasting more than 8 weeks remained positive and significant, after confounding factors (Table 6).

DISCUSSION

The findings of the present study highlight that specific symptoms occurring after COVID-19 infection are associated with the new onset of SUI, UUI, and MUI and may increase the likelihood

Table 2: Variables that were prognostic of urinary symptoms.

Post-COVID-19 symptoms	Wald	df	p-value	Exp(B)	95% CI to Exp(B)	
					Lower	Upper
SUI						
(Constant)	124.99	1	0.01	0.23		
Cough	5.28	1	0.02	1.53	1.06	2.19
Muscle pain	4.40	1	0.04	1.41	1.02	1.95
Loss of movement	4.85	1	0.03	2.15	1.09	4.25
UUI						
(Constant)	239.70	1	0.01	0.18		
Cough	4.34	1	0.04	0.58	0.35	0.97
Diarrhea	5.45	1	0.02	1.75	1.09	2.79
MUI						
(Constant)	105.33	1	0.01	0.16		
Breathing difficulty	5.70	1	0.02	1.81	1.11	2.96
Muscle pain	9.58	1	0.01	2.00	1.29	3.11
Tingling	13.12	1	0.01	2.36	1.48	3.75
Diarrhea	8.83	1	0.01	2.20	1.31	3.70

The binary logistic regression model included all factors which were significant in the univariate analysis using the forward stepwise method. MUI: mixed urinary incontinence, SUI: stress urinary incontinence, UUI: urgency urinary incontinence.

Table 3: Effect of post-COVID-19 symptoms on the quality of life of women with UI.

Post-COVID-19 symptoms	Standardized coefficients	p-value	95% CI		R ²	Difference R ²
	Beta		Lower limit	Upper limit		
(Constant)	-		2.60	3.72	-	-
Muscle pain	0.17	0.01	1.14	2.71	0.06	-
Diarrhea	0.15	0.01	1.27	3.31	0.09	0.03
Tingling	0.12	0.01	0.62	2.39	0.10	0.02
Breathing difficulty	0.1	0.01	0.49	2.33	0.11	0.01

h explains 2.7% of the impact on the t women.

ge, this is the first study that aimed to investigate which symptoms present after COVID-19 infection could explain the UI onset. Previous studies only reported the impact of the COVID-19 pandemic on the presence^{15,16} or worsening of UI¹⁷, however, authors did not describe any odds or risks in reporting UI after the infection. Therefore, our results may redirect the urogynecological management of COVID-19 survivors, as it seems that the presence of specific symptoms after the infection may be associated with UI. Moreover, it is known that UI affects the quality of life of women⁸, therefore, health professionals should be aware about the likelihood of female survivors in developing urinary symptoms.

The primary clinical symptoms of COVID-19 are related to the respiratory system¹⁸, and the greater the pulmonary impairment, more severe is the patient's infection¹⁹. The present study identified an association between respiratory symptoms after COVID-19 infection (i.e., cough and difficulty in breathing) and the new onset of UI (i.e., for SUI, UUI, MUI), as well as adverse impacts on the quality of life of incontinent women. There is a hypothesis that respiratory symptoms may impact urological functionality, as both systems are related⁶. One possible explanation for this association could be the PFM hypoactivity, which might be linked to either muscle weakness or residual respiratory symptoms from COVID-19⁴.

tivation can affect muscle contraction capacity and the maintenance of urinary continence, leading to involuntary loss of urine. In addition, repetitive trauma to the PFM caused by increased intra-abdominal pressure associated with chronic cough may be related to SUI²⁰.

Regarding UUI, a previous study indicated that respiratory disorders have a close relationship with urine storage dysfunction²¹. In addition, it has been reported that COVID-19 infection may be associated with viral cystitis^{22,23}, which could lead to an increased in urinary frequency after recovery from the infection. Therefore, health professionals should also be aware of UUI

Table 6: Correlation between the quality of life of women with UI and symptoms of COVID-19.

ICIQ score	Age & BMI & Parturition
Cough	0.09*
> eight weeks	0.13*
three weeks	-0.02
Breathing difficulty	0.14*
Fatigue	0.17*
Muscle pain	0.22*
Loss of movement	0.12*
Tingling	0.19*
Diarrhea	0.20*

BMI: body mass index; * p-value<0.01

Table 4: Effect of coughing duration on the quality of life of women with UI.

Coughing time	Standardized coefficients	t	p-value	95% CI		R Adjusted ²
	Beta			Lower limit	Upper limit	
(Constant)	-	11.654	0.01	3.45	4.83	-
> eight weeks	0.200	5.042	0.01	1.53	3.49	0.02
three weeks	0.084	2.114	0.03	0.07	1.89	0.03

Table 5: Characteristics of confounding factors according to the presence of urinary incontinence.

3IQ		SUI		UUI		MUI	
		Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Age	Without UI	40.049	9.61	40.46	9.49	38.69	9.39
	With UI	41.932	8.45	40.68	8.73	42.83	10.19
BMI	Without UI	29.9222	6.71	29.87	6.40	28.92	6.13
	With UI	30.4494	6.168	31.00	7.45	31.58	7.01
Parturition	Without UI	1.65	1.18	1.74	1.18	1.53	1.17
	With UI	1.94	1.14	1.64	1.16	1.97	1.20

er risk of reporting SUI was associated after COVID-19 infection. This ent in the musculoskeletal system, which also affects the PFM and continence mechanism. In addition, UI and mobility limitation are associated conditions²⁴ that may result from the interaction of the functions of the trunk, pelvic floor, and lower limb muscles, which together provide orthostatic balance²⁵.

Another interesting result highlights the relationship between diarrhea and UI. Women who experienced diarrhea after COVID-19 infection have 1.75 and 2.2. higher likelihood of reporting UUI and MUI, respectively. However, the relationship between urinary symptoms and diarrhea is still unclear. Experimental studies demonstrated a joint innervation between the colon and the bladder that could lead to cross-organ sensitization²⁶. Moreover, it is already described the association between symptoms of UUI and irritable bowel syndrome²⁷. However, further investigation about the relationship between bowel and bladder symptoms and their impact on women who have had COVID-19 is required.

Neurological symptoms, such as tingling, have been reported as one of the complications of COVID-19 infection since the beginning of the pandemic¹¹. This sensory symptom must be investigated to determine whether there is any sensory deficit that compromises muscle innervation²⁸, which can affect bladder and sphincter function²⁹, potentially leading to symptoms of UI.

Quality of life is defined as a state of complete physical, mental and social well-being³⁰. As a measure of the population's health perception, it is an important aspect to be evaluated in women with UI, specially because even in mild conditions of urinary symptoms, a decline in quality of life can be significant³¹. This study highlights that symptoms identified after COVID-19 infection, such as muscle pain, diarrhea, tingling, difficulty in breathing,

Therefore, health professionals focus on the treatment and prevention of these symptoms in survivors of the COVID-19 infection to minimize their impact on the quality of life of these women.

The present study has some limitations. The first one is related to the study design. As we decided to conduct a cross-sectional study, we could not investigate the causality of the variables. In addition, we could not measure the severity of UI and post-COVID symptoms, which limited our analysis about the association of symptom severity with the onset of UI. Another limitation is related to the data collection period. When the study began, the vaccination program had already been initiated in Brazil. Therefore, we believe that the immunization level of the participants might be a covariable that should have been included in our analysis, as it could have impacted the severity of COVID-19 symptoms. Moreover, the study was limited in identifying whether participants had a chronic cough or were obese prior to COVID-19 infection. Future studies should investigate the relationship between the COVID-19 infection and UI to provide a comprehensive understanding of how UI may worsen in women who were already incontinent before the pandemic. Nonetheless, studies should examine the improvement or worsening of UI following infection through a longer cohort study.

Conclusion

Specific symptoms present after COVID-19 infection can predict SUI, UUI, and MUI. In addition, it seems that muscle pain, diarrhea, tingling, difficulty in breathing and presence of chronic cough can worsen women's quality of life. These findings highlight to the health community the need to provide targeted urogynecological assistance to female survivors of the disease, particularly physiotherapists working with strategies to prevent and treat urinary disorders.

REFERENCES

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